Math140C - HW #4

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1.

Let $\mathcal{M}$ be an algebra on a set X.

(a)

Show $\emptyset, X \in \mathcal{M}$

. . . .

For any $A \in \mathcal{M}$, $A \cup (X \setminus A) = X \in \mathcal{M}$. And hence $X \setminus X = \emptyset \in \mathcal{M}$.

(b)

Show that if $A_1, \ldots, A_n \in \mathcal{M}$, then $\bigcap_{i=1}^n A_i$. Similarly, supposing $\mathcal{M}$ is a σ-algebra, show that if $\{A_n\}_{n \in \mathbb{Z}_+} \subset \mathcal{M}$, then $\bigcap_{i=1}^\infty A_i \in \mathcal{M}$.

. . . .

For the first part, it clearly suffices to show that if $A, B \in \mathcal{M}$, then $A \cap B \in \mathcal{M}$. Well, deMorgan's law says

$$X \setminus ((X \setminus A) \cup (X \setminus B)) = A \cap B,$$

and the left hand side is in $\mathcal{M}$, which shows $A \cap B \in \mathcal{M}$.

For σ-algebras, once again, use deMorgan's Law:

$$X \setminus \bigcup_{i=1}^\infty (X \setminus A_i) = \bigcap_{i=1}^\infty A_i$$

If $\{A_i\}_{i \in \mathbb{Z}_+} \subset \mathcal{M}$, then the left hand side is in $\mathcal{M}$, so $\bigcap_i A_i \in \mathcal{M}$.

2.

(a)

Let I be any set and $\mathcal{M}_i$ a σ-algebra on X for every $i \in I$. Prove $\bigcap_{i \in I} \mathcal{M}_i$ is a σ-algebra on X.

. . . .

Let $\mathcal{M} = \bigcap_{i \in I} \mathcal{M}_i$. Take $A \in \mathcal{M}$. For each $i \in I$, $A \in \mathcal{M}_i$, so it follows $X \setminus A \in \mathcal{M}_i$. This holds for each $i \in I$, so $X \setminus A \in \mathcal{M}$. Closure under arbitrary unions holds by the same line of reasoning.

(b)

Take any $\mathscr{C} \subset \mathscr{P}(X)$. Show there is a unique smallest σ-algebra $\mathcal{M}$ on X that contains $\mathscr{C}$.

. . . .

Let $\{\mathcal{M}_i\}_{i\in I}$ be the collection of all σ-algebras that contain $\mathscr{C}$. Then

$$\bigcap_{i\in I} \mathcal{M}_i$$

is clearly the unique smallest σ-algebra that contains $\mathscr{C}$.

3.

(a)

Let $\mathscr{C}_1, \mathscr{C}_2 \subset \mathscr{P}(X)$. Show $\mathscr{C}_1 \subset \sigma\text{-alg}(\mathscr{C}_2) \iff \sigma\text{-alg}(\mathscr{C}_1) \subset \sigma\text{-alg}(\mathscr{C}_2)$

. . . .

The reverse direction is clear since $\mathscr{C}_1 \subset \sigma\text{-alg}(\mathscr{C}_1)$. The forward direction is also immediate because $\sigma\text{-alg}(\mathscr{C}_1)$ is by definition the smallest σ-algebra containing $\mathscr{C}_1$, hence $\mathscr{C}_1 \subset \sigma\text{-alg}(\mathscr{C}_2)$ implies $\sigma\text{-alg}(\mathscr{C}_1) \subset \sigma\text{-alg}(\mathscr{C}_2)$ by definition.

(b)

Let $\mathcal{I} = \{(a,b) \subset \mathbb{R} \mid a < b \text{ and } a, b \in \mathbb{Q}\}$. Explain why $\mathcal{B}_{\mathbb{R}} = \sigma\text{-alg}(\mathcal{I})$.

. . . .

$\mathcal{I}$ is a subset of all the open sets of $\mathbb{R}$, so $\sigma\text{-alg}(\mathcal{I}) \subset \mathcal{B}_{\mathbb{R}}$ is immediate. For the reverse inclusion, observe that open subsets in $\mathbb{R}$ form a subset of $\sigma\text{-alg}(\mathcal{I})$: suppose $U \subset \mathbb{R}$ is open and pick any $x \in U$. By definition, there exists $r > 0$ such that $\mathrm{B}_r(x) \subset U$. Since $\mathbb{Q}$ is dense in $\mathbb{R}$, one can choose $a, b \in \mathbb{Q}$ such that $(a,b) \subset \mathrm{B}_r(x) \subset U$. Hence U is a union intervals with rational endpoints, i.e.,

$$U \in \sigma\text{-alg}(\mathcal{I}).$$

Since U is an arbitrary open set in $\mathbb{R}$, the above shows the set of all open sets in $\mathbb{R}$ is a subset of $\sigma\text{-alg}(\mathcal{I})$. $\mathcal{B}_{\mathbb{R}} \subset \sigma\text{-alg}(\mathcal{I})$ follows by part (a).

If one knows a bit of topology, this problem is immediate from the fact that open intervals with rational endpoints form a basis for the regular topology of $\mathbb{R}$.

4.

Let $\mathscr{C} = \{\{x\} \mid x \in \mathbb{R}\}$ be the collection of singleton subsets of $\mathbb{R}$. Find a characterization for the sets belonging to $\sigma\text{-alg}(\mathscr{C})$. Specifically, find an explicit condition, and show that the collection of sets satisfying your condition

- includes every set in $\mathscr{C}$
- is a σ-algebra

- is contained in σ-alg($\mathscr{C}$).

. . . .

Note that the outlined conditions indeed show such a collection is σ-alg($\mathscr{C}$). Well, I claim that $\mathscr{P}(\mathbb{R}) = \sigma$-alg($\mathscr{C}$). It is trivially true that $\mathscr{P}(\mathbb{R})$ contains all singletons in $\mathbb{R}$, i.e., every set in $\mathscr{C}$, and is a σ-algebra. To see that $\mathscr{P}(\mathbb{R}) \subset \sigma$-alg($\mathscr{C}$), take any $S \subset \mathbb{R}$. By the definition of a set,

$$S = \bigcup_{x \in S} \{x\},$$

hence $S \in \sigma$-alg($\mathscr{C}$). This proves $\mathscr{P}(R) = \sigma$-alg($\mathscr{C}$).

5.

Let $R = \prod_{i=1}^{p} I_i \in \mathcal{R}$ where each $I_i \subset \mathbb{R}$ is connected. A coordinate-wise partition of R is a partition P of R such that there exists finite partitions Q_i of I_i for each $i \in \{1, \ldots, p\}$ such that

- for each $i \in \{1, \ldots, p\}$, each set in Q_i is connected
- $P = \{\prod_{i=1}^{p} J_i \mid J_i \in Q_i \text{ for each } i \in \{1, \ldots, p\}\}$

(a)

Show that if P is a coordinate-wise partition, then

$$m(R) = \sum_{D \in P} m(D) = \sum_{J_1 \in Q_1} \sum_{J_2 \in Q_2} \cdots \sum_{J_p \in Q_p} m(\prod_{i=1}^{p} J_i). \tag{1}$$

. . . .

After observing that each element of P is itself a bounded rectangle, the second equality in (1) holds by definition, so it suffices to show the equality between the first and third expressions in (1). Let a_i, b_i be for each $i \in \{1, \ldots, p\}$ such that

$$\prod_{i=1}^{p}(a_i, b_i) \subset R \subset \prod_{i=1}^{p}[a_i, b_i]$$
$$m(R) = \prod_{i=1}^{p}(b_i - a_i)$$

Consider any index $i \in \{1, \ldots, p\}$. Without loss of generality, suppose $I_i = [a_i, b_i]$ (in the treatment below, I will write all segments as closed, just for notational simplicity; the distinction of a segment being closed is unimportant for part (a)). Let Q_i be as defined in the problem statement. By the restrictions on Q_i, there exist $x_{i0}, x_{i1}, \ldots, x_{it_i}$ for some integer t_i such that $x_{i0} = a_i$, $x_{it_i} = b_i$, and

$$Q_i = \{[x_{i0}, x_{i1}], [x_{i1}, x_{i2}], \ldots, [x_{i(t_i-1)}, x_{it_i}]\}.$$

By definition of the x_{ij}'s,

$$b_i - a_i = \sum_{j=1}^{t_i}(x_{ij} - x_{i(j-1)}).$$

With t_i and x_{ij} defined as above for all $i \in \{1, \ldots, p\}$ and $j \in \{1, \ldots, t_i\}$,

$$\begin{aligned} m(R) = \prod_{i=1}^{p}(b_i - a_i) &= \prod_{i=1}^{p}(\sum_{j=1}^{t_i}(x_{ij} - x_{i(j-1)})) \\ &= \sum_{j_i=1}^{t_1}\sum_{j_2=1}^{t_2}\cdots\sum_{j_p=1}^{t_p}\prod_{k=1}^{p}(x_{kj_i} - x_{k(j_i-1)}) \qquad (2) \\ &= \sum_{J_1\in Q_1}\sum_{J_2\in Q_2}\cdots\sum_{J_p\in Q_p} m(\prod_{k=1}^{p} J_k), \end{aligned}$$

where (2) is due to the distributive property.

(b)

Show that if $R = \bigsqcup_{k=1}^{n} A_k$ for disjoint squares $A_1, \ldots, A_n \in \mathcal{R}$, then for every $k \in \{1, \ldots, n\}$, there is a coordinate-wise partition P_k of A_k such that $P = \bigcup_{k=1}^{n} P_k$ is a coordinate-wise partition of R.

....

Associate with each A_k points a_{ki}, b_{ki} for all $i \in \{1, \ldots, p\}$ such that

$$\prod_{i=1}^{p}(a_{ki}, b_{ki}) \subset A_k \subset \prod_{i=1}^{p}[a_{ki}, b_{ki}]$$

For each $i \in \{1, \ldots, p\}$, define

$$T_i = \{a_{ki} \mid k \in \{1, \ldots, n\}\} \cup \{b_{ki} \mid k \in \{1, \ldots, n\}\}$$

Clearly, T_i is a finite collection of points lying in the segment $[a_i, b_i]$. Ordering the points of T_i under $<$ as in $T_i = \{p_0, p_1, \ldots, p_{t_i}\}$ for some t_i, each $p_j < p_{j+1}$, let

$$Q_i = \{(p_0, p_1), (p_1, p_2), \ldots, (p_{t_i-2}, p_{t_i-1}), (p_{t_i-1}, p_{t_i})\} \cup \bigcup_{j=0}^{t}\{p_j\}$$

By construction, the Q_i's are a finite partition of the corresponding I_i's such that

$$P = \left\{\prod_{i=1}^{p} J_i \mid J_i \in Q_i \text{ for each } i \in \{1, \ldots, p\}\right\}$$

is a coordinate-wise partition of R. Crucially, P induces a coordinate-wise partition for each A_k. For a fixed dimension $i \in \{1, \ldots, p\}$ and fixed rectangle A_k for $k \in \{1, \ldots, n\}$, one can form the segment I_{ki} defining the ith dimension of A_k by taking union of elements of Q_i; namely, that the endpoints of I_{ki} are added to Q_i as singletons means the elements of Q_i can form open, closed, and half-open intervals with endpoints a_{ki}, b_{ki}. In other words, given a rectangle A_k, each Q_i admits a subset partitioning I_{ki} that defines the coordinate-wise partition for A_k. Furthermore, it is clear that the only shared points among the P_k's are rectangles of measure zero; in each coordinate, the segments $I_{k_1 i}$ and $I_{k_2 i}$ for $k_1 \neq k_2 \in \{1, \ldots, n\}$ share at most their endpoints. Conversely, all elements of P belong to a coordinate-wise partition of at least one of the A_k's.

Writing P_k as this induced coordinate-wise partition of A_k, the conclusion obtained from the above paragraph is that $P = \bigcup_{k=1}^n P_k$.

(c)

Let the notation be as in part (b). Combining the results from parts (a) and (b),

$$\begin{aligned} m(R) = \sum_{D \in P} m(R) &= \sum_{k=1}^{n} \sum_{D \in P_k} m(D) \\ &= \sum_{k=1}^{n} m(A_k), \end{aligned}$$

as was to be shown.

6.

(a)

Let $R \in \mathcal{R}$ and let $R_1, \ldots, R_n \in \mathcal{R}$ be disjoint subsets of R. Show there is a coordinate-wise partition P of R so that each R_k is a union of sets from P. Conclude that $R \setminus (R_1 \cup \ldots \cup R_n)$ can be expressed as a finite union of disjoint rectangles.

. . . .

For the first part of this question, proceed more or less exactly as in Q5(b). The only thing to note is that excluding sets of P of measure 0, each R_k is a disjoint union of sets of P; since the R_k's are themselves disjoint, the sets of P used to build the rectangles R_k contribute to only one R_k if they have measure greater than 0. Since P partitions R into subrectangles, and P induces further coordinate-wise partitions of the R_k's, $R \setminus (R_1 \cup \ldots \cup R_n)$ can be expressed as a union of disjoint rectangles.

(b)

Prove that every $A \in \mathscr{E}$ can be expressed as a finite union of disjoint rectangles.

. . . .

By definition, $A = \bigcup_{k=1}^n A_k$ for some integer n, $A_k \in \mathcal{R}$, where the A_k's may not be pairwise disjoint. To show that any $A \in \mathcal{E}$ can be expressed as the union of disjoint rectangles, induct on n. The $n = 1$ is trivially true. Consider $n > 1$ with the proposition true for $n - 1$. Rewrite

$$A = \bigcup_{k=1}^{n-1} A_k \cup A_n$$

$A' := \bigcup_{k=1}^{n-1} A_k$ is an element of $\mathscr{E}$, so by the inductive hypothesis, A' can be expressed as the union of disjoint rectangles. If A' is disjoint from A_n, then A is the union of pairwise disjoint rectangles. Suppose A' intersects A_n; particularly, there are some number of disjoint rectangles $B_1, \dots, B_t$ of A' that intersect A_n. Clearly, each $B_j \cap A_n$ is a rectangle subset of A_n, hence part (a) says that

$$A_n \setminus (A_n \cap \bigcup_{j=1}^{t} B_t)$$

can be written as a union of disjoint rectangles. In other words, part of A_n that doesn't intersect A' can be writ as a union of disjoint rectangles. Hence A can be writ as a union of disjoint rectangles.

(c)

Let $A \in \mathscr{E}$ and $R_1, \dots, R_k \in \mathcal{R}$ be one set of pairwise disjoint rectangles, $S_1, \dots, S_l \in \mathcal{R}$ another set of pairwise disjoint rectangles, and $A = \bigsqcup_{i=1}^{k} R_i = \bigsqcup_{j=1}^{k} S_j$. Show $\sum_{i=1}^{k} m(R_i) = \sum_{j=1}^{l} m(S_j)$.

. . . .

For any $i \in \{1, \dots, k\}$, $R_i \subset A = \bigsqcup_{j=1}^{l} S_j$ means

$$\begin{aligned} R_i &= R_i \cap \bigsqcup_{j=1}^{l} S_j \\ &= \bigsqcup_{j=1}^{l} (R_i \cap S_j) \qquad (3) \end{aligned}$$

(and indeed, simple set theory shows (3) is still a disjoint union). The intersection of rectangles is clearly another rectangle; (3) shows the rectangle R_i is a disjoint union of rectangles. Hence Q5(c) applies to give

$$m(R_i) = \sum_{j=1}^{l} m(R_i \cap S_j)$$

Not only is the choice of i arbitrary, but also a symmetric analysis can be to write each S_j as a disjoint union of rectangles, yielding

$$m(S_j) = \sum_{i=1}^{k} m(S_j \cap R_i).$$

Putting these results together,

$$\begin{aligned} \sum_{i=1}^{k} m(R_i) &= \sum_{i=1}^{k} \sum_{j=1}^{l} m(R_i \cap S_j) \\ &= \sum_{j=1}^{l} \sum_{i=1}^{k} m(R_i \cap S_j) = \sum_{j=1}^{l} m(S_j), \end{aligned}$$

as was to be shown.